Preg Assist

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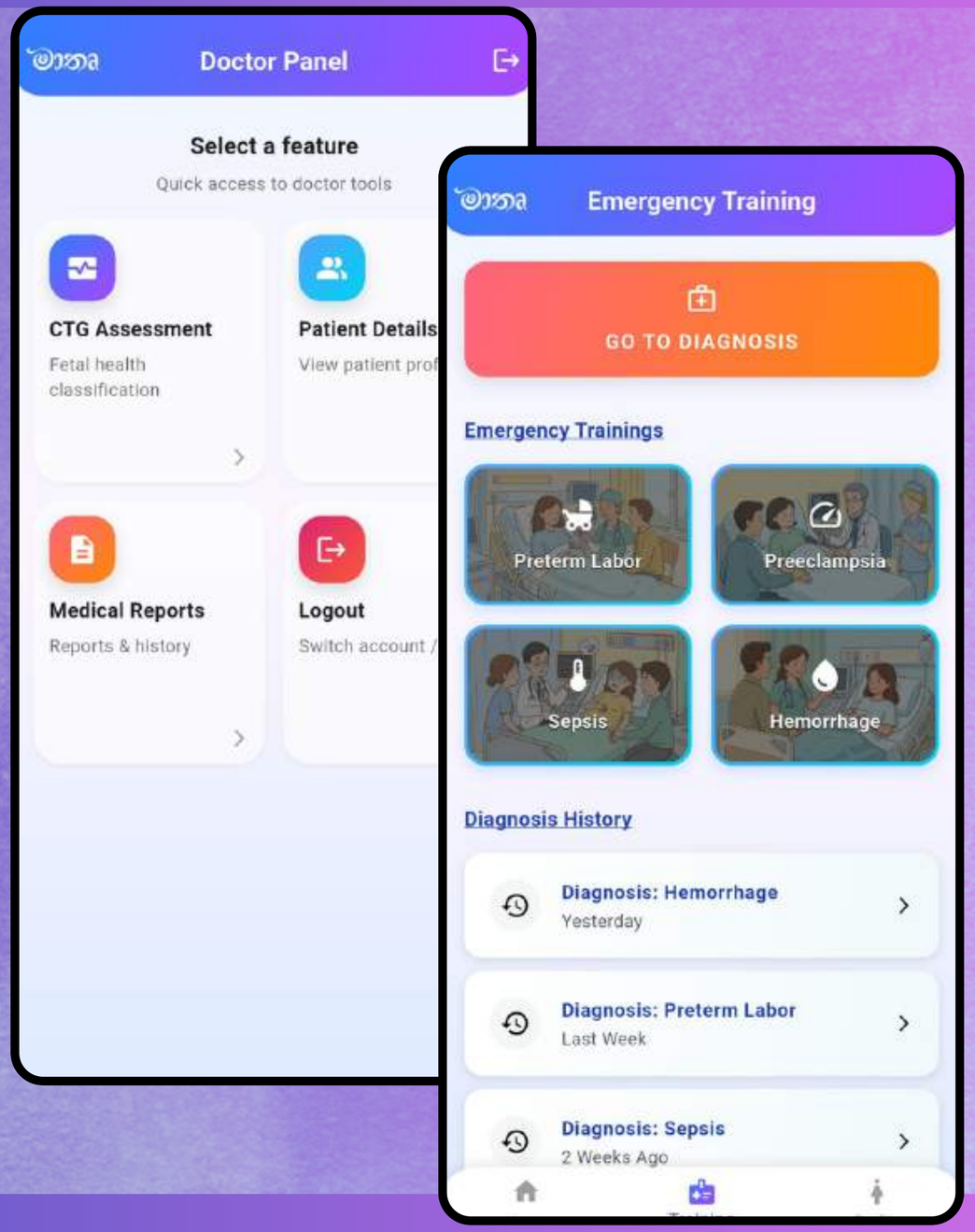

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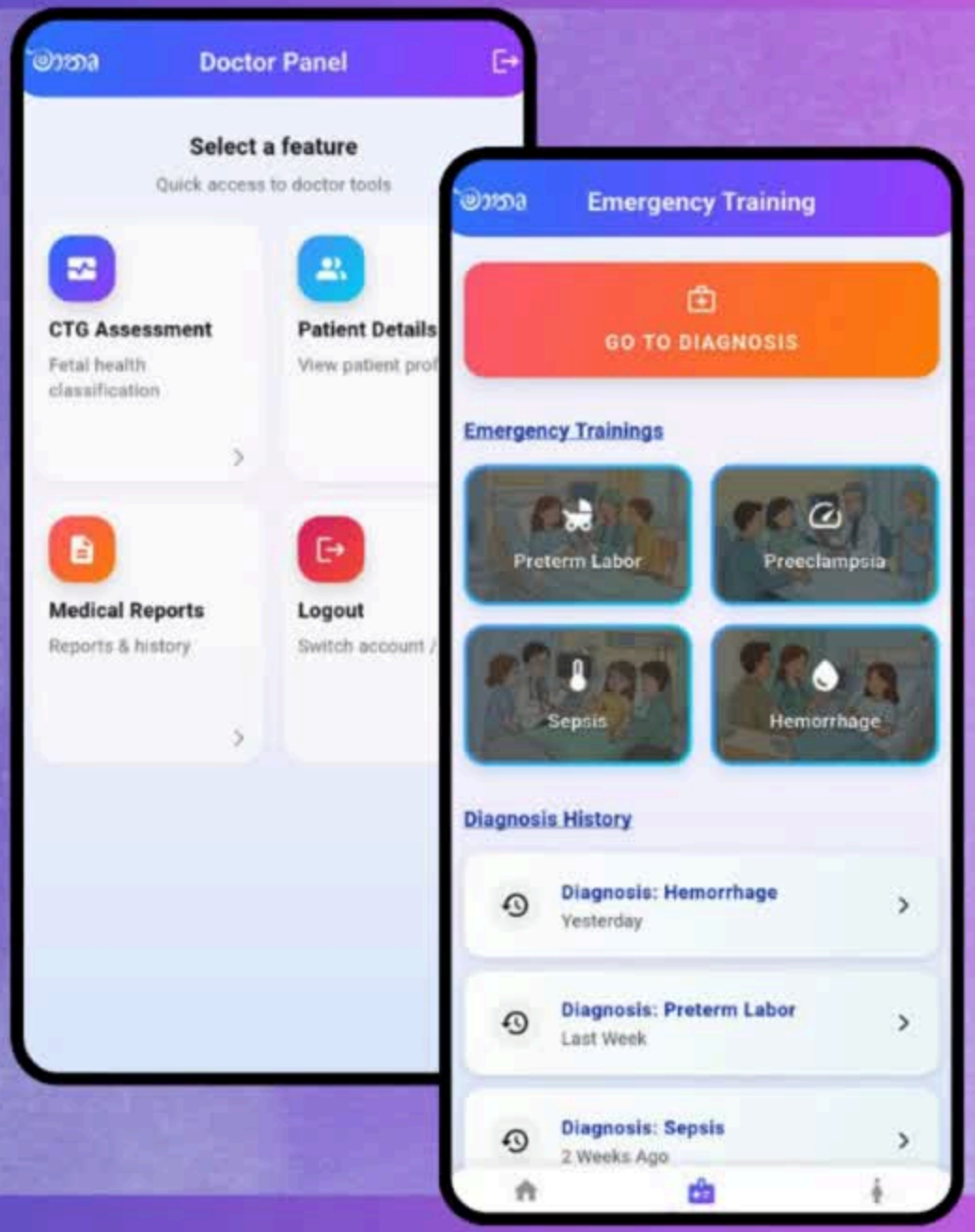

Problem & Its Impact / Gravity

Problem

Pregnancy requires continuous monitoring of maternal physical health, mental wellbeing, and fetal condition, but current care is mostly limited to clinic visits. Important risks such as high blood pressure, blood sugar, emotional stress, and fetal distress are often identified late. Existing digital and AI-based solutions typically focus on isolated medical data and provide limited support for maternal mental, health and education. As a result, pregnant mothers often lack clear guidance, emotional support, and understanding of fetal development. Therefore, an integrated intelligent prenatal care system is needed to combine health monitoring, mental health support, educational guidance, and clinical decision support in one accessible platform

Impact / Gravity

1. Reduction of maternal and fetal risks
2. Improved maternal mental wellbeing
3. Enhanced clinical decision-making
4. Increased maternal health literacy and engagement
5. Accessibility in low-resource settings

Who Will Benefit

Affected Groups

- VOG Clinicians
- Pregnant Mothers

Institutions / Society

- Hospitals and maternity clinics
- Pregnant mothers and families
- Rural and underserved communities
- Infants and future generations

Beneficiaries of the Solution

- Pregnant mothers – Improved health monitoring, mental wellbeing support, and pregnancy education.
- Obstetricians and gynecologists – Explainable decision support for accurate fetal health assessment.
- Healthcare workers in resource-limited settings – Clinical decision support for fetal health assessment with reduced dependence on continuous internet connectivity.
- Healthcare institutions – Reduced maternal and fetal risks and improved efficiency in prenatal care services.

Design Excellence of the Overall Solution

Integrated and Modular Architecture

Design Excellence: Unified yet independent system design

- The solution is designed as four tightly integrated but independent modules: education, maternal physical health, mental health, and fetal decision support.
- Each module addresses a distinct clinical or educational gap while sharing a common data and workflow backbone.
- This modular design allows components to function independently or together, ensuring system robustness and flexibility.
- The architecture supports future expansion without disrupting existing components, making the solution scalable and maintainable.

Why this is excellent:

It avoids fragmented solutions while preventing monolithic complexity.

Design Excellence of the Overall Solution

User-Centered Design for Dual Stakeholders

Design Excellence: Tailored experiences for mothers and clinicians

- The system is designed for two primary user groups: pregnant mothers and healthcare professionals.
- Mothers receive simplified guidance, educational support, and alerts without clinical complexity.
- Clinicians receive structured health summaries, fetal classification results, and explainable insights for decision support.
- This separation of interaction layers reduces cognitive overload and improves trust and usability for both users.

Why this is excellent:

It aligns technical outputs with real-world user needs and clinical workflows.

Design Excellence of the Overall Solution

Privacy-Preserving and Trust-Focused Design

Design Excellence: Sensitive data handled responsibly

- Mental health analysis and emotion detection are designed to run on-device, minimizing exposure of sensitive data.
- Educational and monitoring components avoid unnecessary data sharing, reducing privacy risks associated with online platforms.
- Federated or adaptive learning strategies enable system improvement without transmitting raw personal data.
- The design prioritizes user trust, a critical requirement in maternal and mental healthcare systems.

Why this is excellent:

Privacy is treated as a core design principle, not an afterthought.

Design Excellence of the Overall Solution

Explainable and Resource-Efficient Intelligence

Design Excellence: Practical AI for real-world healthcare

- The fetal health DSS emphasizes explainable predictions, allowing clinicians to understand why a decision is made.
- Models are optimized for mobile and edge deployment, balancing accuracy with latency, storage, and power constraints.
- Multimodal mental health analysis combines text, voice, and image cues to improve prediction reliability.
- The system avoids black-box intelligence and instead delivers actionable, interpretable outputs.

Why this is excellent:

It ensures AI is usable, trustworthy, and deployable in practical healthcare settings.

Decision Support System for Fetal Health Classification

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Information Technology

Decision Support System for Fetal Health Classification

How the Solution will address the Sub-Problem / Prototype

- Provides consistent decision support during and beyond clinic visits, improving assessment continuity.
- Assists obstetricians and gynecologists by confirming and strengthening clinical judgments in critical situations.
- Generates explainable outputs that highlight feature contributions, enabling transparent and informed decisions.
- Supports early identification of abnormal fetal conditions, helping prevent adverse pregnancy outcomes.

User Requirements addressed by the solution

- **Need for accurate and reliable fetal health assessment** → addressed through machine learning–based classification.
- **Need for decision transparency and trust** → addressed through explainable predictions and feature importance.
- **Need for fast and practical clinical support** → addressed through lightweight, mobile-ready DSS design.
- **Need to reduce cognitive burden during CTG interpretation** → addressed by structured, automated insights.
- **Need for support in resource-limited or time-constrained settings** → addressed through optimized on-device inference.

Decision Support System for Fetal Health Classification

Design Excellence/Contribution

- **Balances model accuracy with on-device efficiency**, enabling fast inference suitable for mobile deployment.
- **Provides clinically meaningful explanations** by highlighting key contributing CTG parameters for each prediction.
- **Designed to support not replace**, clinical judgment, strengthening obstetricians' confidence in critical situations.
- Demonstrates a practical pathway for integrating **transparent AI** into prenatal clinical workflows.

User Feedback on Prototype

Based on pilot demonstrations

- Clinicians found the prediction outputs easy to understand and useful as a second opinion during fetal assessment.
- Users appreciated the clear separation between decision support and final clinical responsibility.
- Feedback highlighted the potential value of the DSS in resource-limited or high-workload environments.
- Shows the potential scalability I can reach by integrating my system to hospital systems.

AR and AI-Driven Module for Personalized Pregnancy Emergency Training

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Information Technology

AR and AI-Driven Module for Personalized Pregnancy Emergency Training

How the Solution will address the Sub-Problem / Prototype

- **Hybrid Architecture:** Couples lightweight Flutter logic with high-fidelity Unity AR.
- **Intelligent Routing:** Deep-links diagnoses directly to specific AR drills (no manual searching).
- **Offline Capability:** Fully functional inference engine and AR assets without internet.
- **Spatial Simulation:** Replaces static text with immersive 3D emergency overlays.

User Requirements addressed by the solution

- **Immediate Triage:** Delivers diagnostic results in less than 1 second via local processing.
- **Trust & Safety:** Deterministic Expert System ensures 100% adherence to WHO guidelines.
- **Low Cognitive Load:** Linear "Wizard" interface reduces panic-induced decision fatigue.
- **Visual Clarity:** Language-independent AR cues guide actions instantly.

AR and AI-Driven Module for Personalized Pregnancy Emergency Training

Design Excellence/Contribution

- **Custom Deep Linking:** Engineered CLEAR_TASK protocol for stable Flutter-to-Unity context switching.
- **Explainable AI:** Prioritized Rule-Based systems over ML to guarantee medical consistency.
- **Cross-Platform Bridge:** Seamless passing of complex drill states (Drill IDs) between Native and Unity.
- **Decoupled Modularity:** Logic and AR assets update independently for easier maintenance.

User Feedback on Prototype

- **Rapid Navigation:** Validated in less than 2 seconds transition time from Diagnosis Result to AR Scene.
- **Reduced Anxiety:** Users preferred the step-by-step wizard over traditional scrollable forms.
- **Better Retention:** AR visualization improved recall of emergency body positioning.
- **Instructional Clarity:** Audio-visual cues in AR were rated clearer than standard text manuals.

AI-Based Mental Health Monitoring and Risk Prediction for Pregnant Women

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Information Technology

AI-Based Mental Health Monitoring and Risk Prediction for Pregnant Women

How the Solution will address the Sub-Problem / Prototype

- **Multimodal Emotion Capture** – Uses facial expressions, voice, and text inputs to assess mental health more accurately.
- **Privacy Preserving Learning** – Uses federated learning to improve model accuracy without sharing raw sensitive data.
- **Emotion Weighted Risk Scoring** – Combines all inputs using a weighted fusion method with higher importance on emotion detection.
- **Adaptive Support and Alerts** – Delivers personalized interventions or triggers trusted contact alerts based on risk level.

User Requirements addressed by the solution

- **Accurate Mental Health Assessment** – Provides a reliable evaluation by analyzing emotions from multiple input types.
- **Strong Privacy Protection** – Ensures user data safety through on device processing and no storage of raw sensitive data.
- **Timely Emotional Support** – Provides immediate personalized support when risk is detected.
- Safety in Critical Situations – Alerts a trusted contact during high mental health risk.

AI-Based Mental Health Monitoring and Risk Prediction for Pregnant Women

Design Excellence/Contribution

- **Emotion Centered Multimodal Design** – Prioritizes emotion detection by combining facial, voice, and text inputs for higher assessment accuracy.
- **Privacy First Architecture** – Ensures sensitive data is processed on-device with no storage of raw images or audio.
- **Federated Learning Design** – Enables model improvement without sharing user data.

User Feedback on Prototype

- Users trusted the system due to strong privacy protection
- Sense of Safety – User felt more secure knowing the system could alert a trusted contact during high-risk emotional situations.
- Users found the system simple and non intrusive, allowing them to share emotional inputs naturally without added stress.

Physical Health Monitoring and Wellness guidance for Pregnant Mothers

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Information Technology

Physical Health Monitoring and Wellness guidance for Pregnant Mothers

How the Solution will address the Sub-Problem / Prototype

- **Smart Risk Detection –** Our AI analyzes all the health vitals at once to find hidden risks that single tests might miss.
- **Trimester-Based Guidance –** The app changes its medical advice based on the pregnancy stage.
- **Clear Health Explanations –** Instead of a random score, the system shows exactly which vital sign caused the risk alert.
- **Emergency Action Alerts –** The app delivers instant, personalized steps to take when a high-risk level is detected to ensure immediate safety.

User Requirements addressed by the solution

- **Comprehensive Health Monitoring –** Our XGBoost AI analyzes multiple vitals together to identify complex risk patterns that single tests often miss.
- **Stage-Specific Medical Guidance –** The app adjusts its risk checks and advice based on the current trimester to watch for specific dangers like Pre-eclampsia.
- **Transparent Risk Explanations –** The system uses SHAP technology to show exactly which vital sign caused a high-risk alert so you understand the result.
- **Real-Time Emergency Alerts .**

Physical Health Monitoring and Wellness guidance for Pregnant Mothers

Design Excellence/Contribution

- **Integrated XAI Framework –** Uses SHAP technology to show the user exactly which vital sign caused their risk alert instead of just giving a random score.
- **Trimester-Aware Logic Engine –** Automatically adjusts risk checks and medical advice based on the pregnancy stage to catch specific dangers like Pre-eclampsia.
- **User-Centric Emergency Tiering –** Provides a clear risk hierarchy that gives immediate, personalized action steps to ensure the mother's safety.
- **Multivariate Pattern Analysis**

User Feedback on Prototype

- Trust in Results – Users liked that the app explained exactly why they were at a certain risk level.
- Helpful Pregnancy Stages – Mothers found the advice very useful because it changed based on their current trimester.
- Easy to Understand – Users felt the "Low, Mid, and High" risk levels were very clear and simple to follow.
- Feeling of Safety – Mothers felt safer knowing the app would warn them immediately about emergencies like Pre-eclampsia.

Preg Assist

What Is Unique in Our Solution

- All-in-one intelligent prenatal care platform
- Explainable AI–based fetal decision support
- Privacy-aware design with on-device intelligence
- Interactive AR-based prenatal education
- Mobile-first and resource-conscious design

How Will the Cost Be Recovered?

- Premium clinical features (advanced analytics, reporting, DSS access) for professionals.
- Partnerships with hospitals
- Freemium access for pregnant mothers, with optional paid value-added services.

Preg Assist

Targeted Customers

- Pregnant mothers seeking reliable health monitoring, mental wellbeing support, and pregnancy education.
- Obstetricians and gynecologists
- Hospitals and maternity clinics aiming to improve prenatal care quality and efficiency

Market Size

- Large and growing prenatal care market, driven by increasing maternal health awareness and digital health adoption.
- High smartphone penetration enables wide reach through mobile-based solutions.
- Demand in both urban and resource-limited settings for scalable prenatal care tools.

Thank You!